

Silicon Photonics Design, Fabrication and Data Analysis edX Course Report

Fedor Goncharov

edX name: feddd0r

Abstract: This report presents the complete photonic integrated circuit (PIC) design cycle, covering component-level simulations, circuit assembly, and fabrication-ready layout generation. Several Mach–Zehnder interferometers and MMI couplers were designed, manufactured, and experimentally characterized, enabling extraction of effective and group refractive indices and assessment of device robustness through corner analysis. While measured parameters deviated slightly from expectations, particularly in effective index and MMI imbalance, the results validate the overall functionality of the devices and provide a foundation for further refinement and performance verification.

1. Introduction

Photonic integrated circuits (PICs) are the promising approach for many applications including optical communications, computation, sensing, and so on [1]. PICs allow to design scalable, stable, small, and ready to mass production solutions. Silicon photonics is one of the technologies commonly used for PICs implementations. It benefits from CMOS capability, which means that it is possible to fabricate such photonic chips using the same equipment as for electronics.

The project objective is to go through all design steps from idea to fabrication, including experimental data analysis. Namely, the focus of the project is Mach-Zehnder interferometer (MZI). MZI is a basic building block for a lot of applications, such as optical switchers, filters, optical processors for neuromorphic neural networks [2], and many others.

In the project, the standard waveguide cross section for Silicon on Insulator (SOI) [3] of 500 x 220 nm, and quasi-TE mode is chosen for work.

2. Theory

2.1. Waveguides

A waveguide is an optical pipe which guides light. The refractive index contrast is required for light guide: the core refractive index has to be higher than the cladding's index. Light propagates through the waveguide as modes. A mode is a stationary field profile of a traveling wave and it is obtained as a wave equation solution for a particular waveguide geometry. Waveguides may be single-mode or multi-mode. The plane wave in a waveguide is given as follows:

$$E = E_0 e^{i(\omega t - \beta z)}$$

with propagation constant β :

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda},$$

where n_{eff} is an effective index and λ is a wavelength. Besides the effective index describing the phase velocity $v_p = c/n_{\text{eff}}$, the group index is needed to describe a group velocity ($v_g = c/n_g$):

$$n_g = n_{\text{eff}} - \lambda \frac{dn}{d\lambda}$$

Group velocity describes the propagation of the entire pulse consisting of multiple wavelengths.

Polarization also plays a role in a waveguide. Two polarization modes are distinguished in a waveguide:

- The transverse electrical (TE) mode with electrical field transverse to the direction of propagation and parallel to the wafer surface;
- The transverse magnetic (TM) polarization with magnetic field being parallel to the wafer surface.

A waveguide cross-section geometry is very important since it determines an effective and group index. Usually, it is preferred to work with a single mode. In a slab waveguide, the effective index dependence on thickness is provided in Figure 1(a). The 220 nm thickness is standard for many fabrications. Then the width is chosen to maximize the effective index while maintaining the single-mode regime. A higher effective index means that more field is focused in the middle of a waveguide. Thus, for TE-mode the 500 nm is a common width (Figure 1(b)).

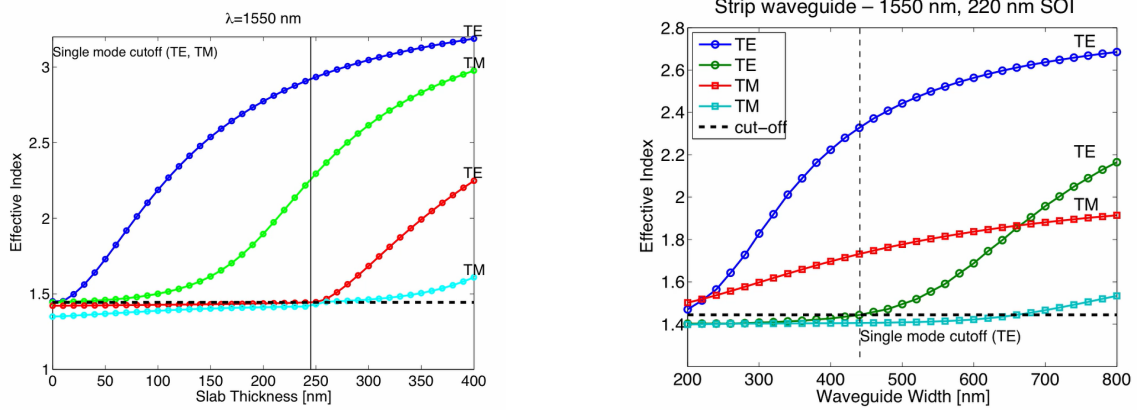


Fig. 1. Effective index dependence on (a) thickness, (b) width

Since core and cladding materials are dispersive media, the effective index is wavelength-dependent. Basically, the effective index can be described in the compact model as a Taylor expansion around the central wavelength, λ_0 :

$$n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2,$$

where n_1 , n_2 , n_3 , λ_0 are the parameters for the waveguide compact model.

The propagation constant with losses is given as

$$\beta(\lambda) = \frac{2\pi n_{eff}(\lambda)}{\lambda[\mu m]} + i \frac{\alpha_{\mu m^{-1}}}{2},$$

where $\alpha_{\mu m^{-1}}$ is the propagation loss per micrometer, and it could be converted from a commonly used $\alpha_{dB/cm}$ as:

$$\alpha_{\mu m^{-1}} = \frac{\alpha_{dB/cm}}{4.34} \cdot 10^{-4}$$

2.2. Waveguide Y-branch

Y-Branch splits one waveguide into two waveguides (a splitter) or vice versa (a combiner). In the case of a splitter, a light input field E_{in} splits equally for two outputs with the fields of $E_{out1} = E_{out2} = E_{in}/\sqrt{2}$. For the combiner case, the output power is the sum of two inputs divided by the square root of two:

$$E_{out} = \frac{E_{in1} + E_{in2}}{\sqrt{2}}$$

The $1/\sqrt{2}$ factor appears due to the existence of at least one more mode except mode in the output waveguide.

2.3. Waveguide bends

The light propagating through the bend is pushed out to the side. Thus, two additional loss mechanisms appear: mode-mismatch and radiational losses. The first one occurs after the bend, since some of the light goes to the higher-order modes. The second one is due to the shifting of light to the border of the waveguide and scattering on its roughness. For a SOI platform, a radius from 5 to 10 microns is sufficient for negligible losses.

2.4. Mach-Zehnder Interferometer

A Mach-Zehnder interferometer (MZI) consists of one splitter, two waveguides, and one combiner. If we consider the Y-branch as a splitter and combiner, the output intensity of the MZI (for the lossless case) will be:

$$I_{out} = \frac{I_{in}}{2}(1 + \cos(\beta_1 L_1 - \beta_2 L_2)),$$

where I_{in} is the input intensity, β_1 and β_2 are the propagation constants of both arms of the waveguides, and L_1 , L_2 are their lengths. If $L_1 = L_2$, the MZI is balanced; if $L_1 \neq L_2$, the MZI is unbalanced.

Generally, the MZI spectrum has a sinusoidal dependence on wavelength. The wavelength distance between two neighbors is given by a free spectral range (FSR). If the arms waveguides are identical (i.e. $\beta_1 = \beta_2$), FSR is given by:

$$\text{FSR} = \frac{\lambda^2}{\Delta L n_g} [nm]$$

For an imbalanced MZI with identical waveguides, the transmission is given by

$$T_{\text{MZI}} = \frac{1}{4} |1 + e^{-i\beta(\lambda)\Delta L}|^2$$

2.5. Multimode interferometer coupler

In addition to MZI, one more device was chosen for fabrication and analysis. A multimode interferometer (MMI) is a device with N input and M output ports. Multimode waveguides are defined as waveguides that support a propagation of more than three modes. Due to multimode interference and the self-imaging principle [4], this device is used for splitting radiation. In this course, I decided to model 2x2 MMI splitter as shown in Figure 2. The geometric parameters are taken from [5] and listed in Table I.

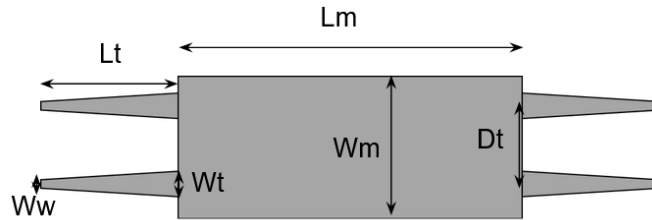


Fig. 2. MMI geometry

3. Modeling

3.1. Waveguide cross-section modeling

First, we start with cross-section analysis. The geometry of the cross section is provided in Figure 3. Simulations are performed in ansys Lumerical MODE [6]. Then, we need to ensure that the

TABLE I
List of fabricated devices and their geometries

MMI parameter	Values [μm]
Wm	3.6
Lm	46.7
Wt	1.5
Lt	25
Dt	2.1
H (y-axis)	0.22
Ww	0.5

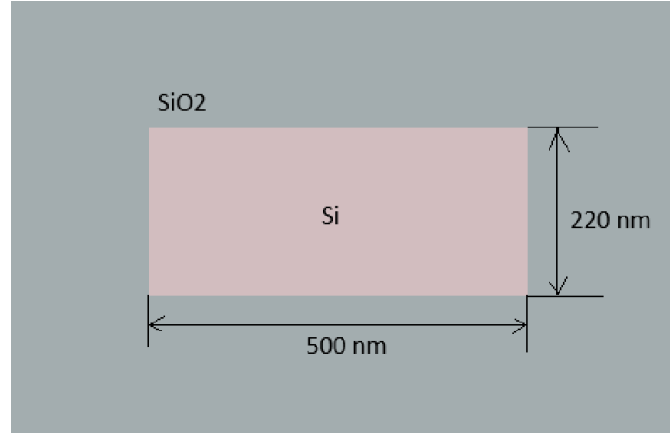


Fig. 3. Cross section geometry of the waveguide

simulation area is enough and that the boundaries do not influence the result. For this purpose, the E-field distribution is estimated on a logarithmic scale, as shown in Figure 4(a). It can be seen that the field is almost decayed in the boundaries. The mesh is small enough: 10 nm for both axes out of the waveguide, 10 and 5 nm inside in y and z directions. Besides, it matches the boundaries of the waveguide. Finally, the fundamental TE-mode field distribution for a wavelength of 1550 nm is obtained and shown in Figure 4(b).

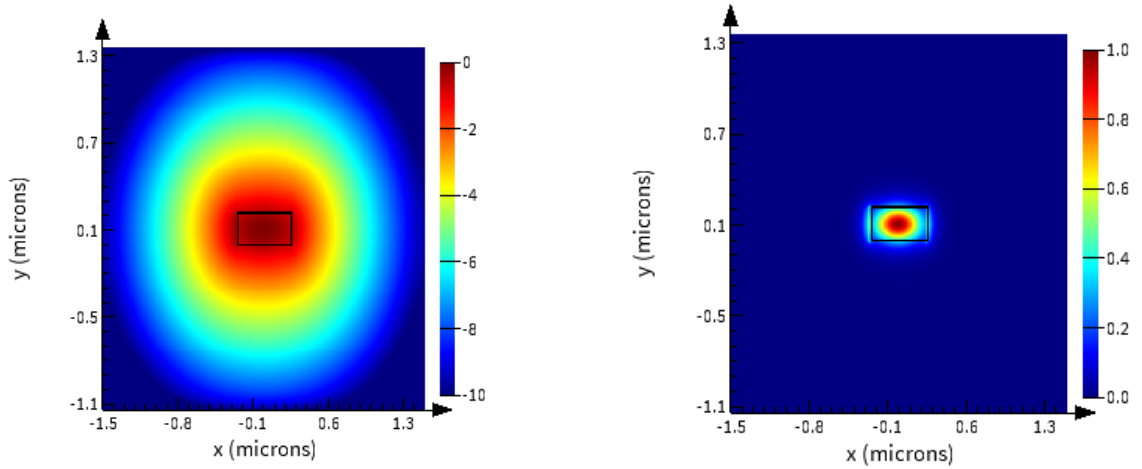


Fig. 4. (a) Field profile in the log scale for boundary influence analysis, (b) Fundamental TE-mode field distribution

Then, the effective and group indices are calculated (Figure 5).

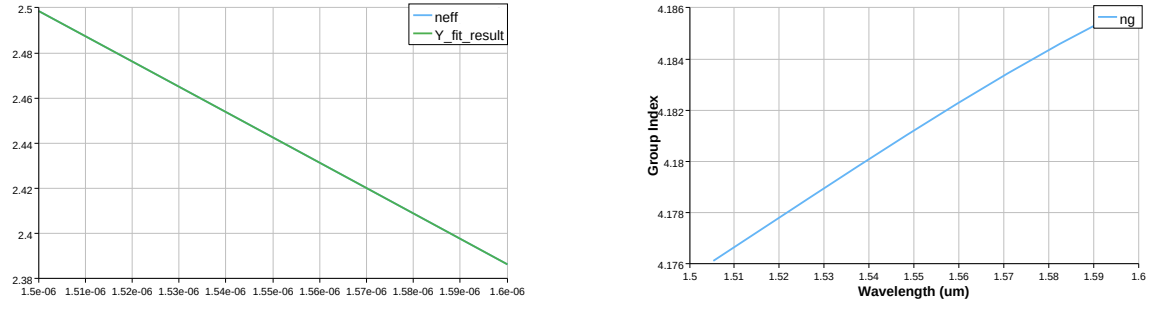


Fig. 5. Effective and group index wavelength dependence

Finally, the compact waveguide model was obtained.

$$n_{\text{eff}} = 2.44237 - 1.1217 * (\lambda - 1.55) - 0.0350225 * (\lambda - 1.55)^2$$

Thus, the group index and dispersion may be easily obtained:

$$n_g = n_1 - n_2 \lambda_0 = 4.1811$$

$$D = -\frac{2\lambda_0 n_3}{c} = 0.00036215$$

3.2. Y-branch and fibre grating coupler modeling

For the MZI design, the y-branch splitter/combiner and fibre grating couplers are needed. In this course, the SiEPIC-EBeam-PDK [7] is used with already prepared element models. Thus, it is possible to simulate their behavior in the circuit level, in the Lumerical INTERCONNECT [8].

Firstly, it is desired to measure spectrum of fibre grating couplers for entering and escaping radiation into and out of the chip (Figure 6).

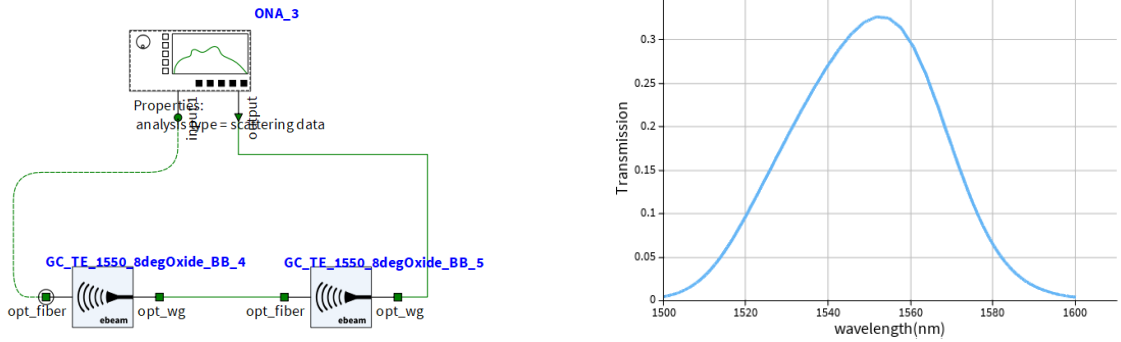


Fig. 6. (a) Scheme for measurement, (b) Transmission spectrum of two grating couplers

Secondly, the y-brach outputs are measured (Figure 7)

3.3. MZI modeling

For MZI design, the aim is to obtain values of ΔL the way to transmit the central wavelength with almost no loss, and investigate the FSR dependence on arm length difference ΔL . Consequently, we need to solve the equation:

$$T_{\text{MZI}}(1.5484[\mu\text{m}]) = 1 = \frac{1}{4} |1 + e^{-i\beta(1.5484[\mu\text{m}])\Delta L}|^2$$

This equation has multiple solutions. Some of them were chosen for the design. (Initially, the wavelength of $1.55 \mu\text{m}$ was chosen for the design. However, after sending the chip for fabrication,

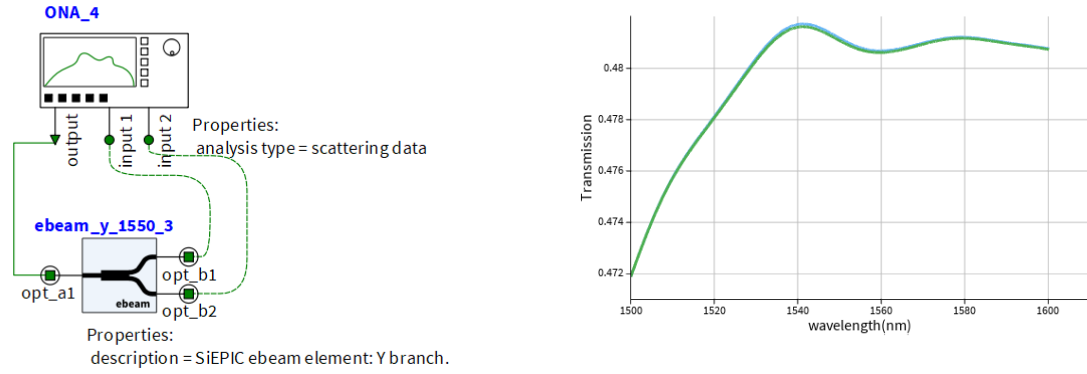


Fig. 7. (a) Scheme for measurement, (b) Transmission spectra of different outputs of Y-branch

I found an error in the waveguide model; thus, I corrected the central wavelength to correct calculations).

Firstly, the simple MZI design, consisting of two standard Y-branches and standard waveguides with parameters from the obtained compact model, was simulated. The losses were considered as 3 dB/cm. The circuit is shown in Figure 8a, while the resulting spectra in Figure 8b.

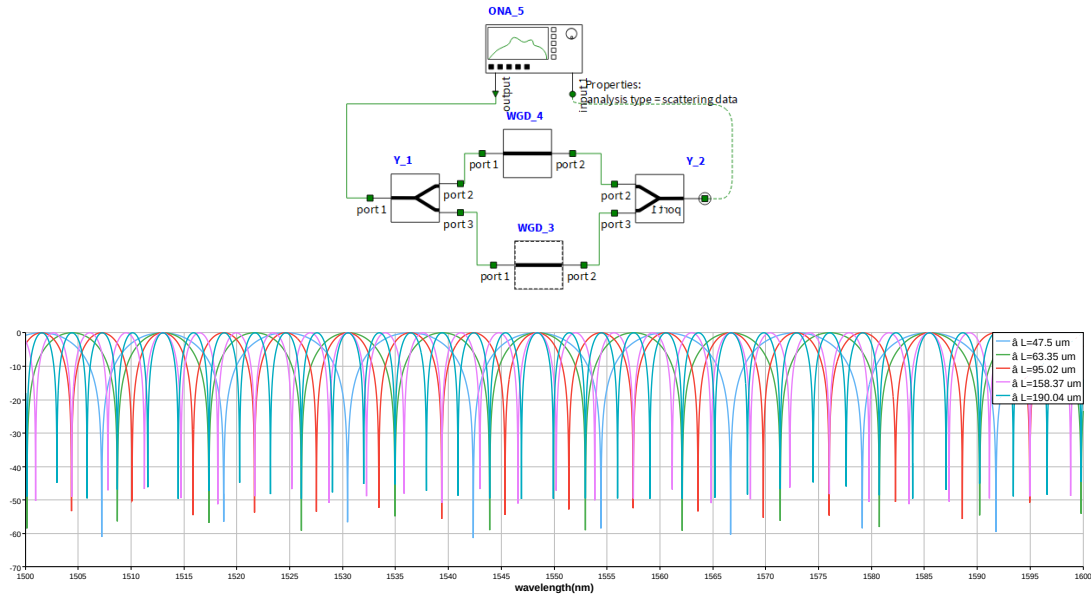


Fig. 8. (a) Scheme for simulation based on standard models and waveguide compact model, (b) Transmission spectra of MZI model based on standard models

Then, the standard models of Y-branches were replaced by ones from PDK, and the grating couplers were added as well. The circuit and the resulting spectra are denoted in Figure 9. Here, it could be clearly seen that the grating coupler transmission is taken into account. The resulting spectra are expected to be close to the experimental raw data.

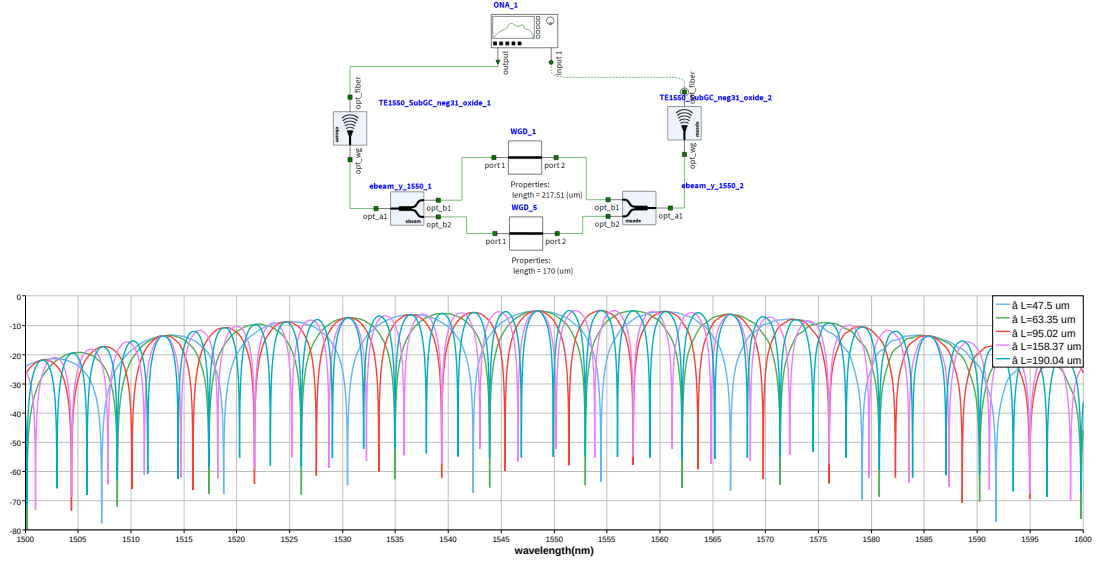


Fig. 9. (a) Scheme for simulation based on PDK models and waveguide compact model, (b) Transmission spectra of MZI model based on PDK models

3.4. MMI modeling

The eigenmode expansion (EME) method is the most suitable for structures like MMI since they are long and have a constant or slowly varying cross-section. First of all, we perform a cross-section analysis of the MMI waveguide to determine if the modeling area is large enough, see Figure 10a. The mesh was selected to be 10 nm inside the MMI waveguide and 50 nm outside it. For correct EME calculation, it is important to analyze enough modes. The mode convergence test was performed. For the MMI region, 18 modes are enough; while for taper region, only 10 modes are needed to converge. These conclusions were made from the mode convergence results as shown in Figures 10 b,c.

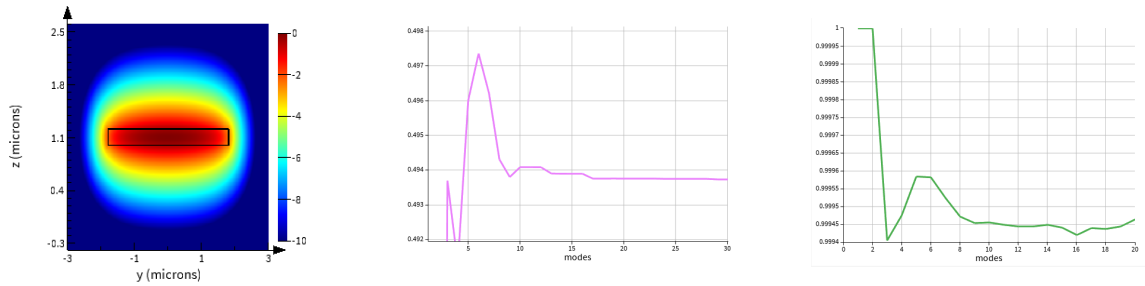


Fig. 10. (a) Cross-section analysis of MMI region in log-scale, (b) Mode convergence for MMI region, (c) Mode convergence for taper

Besides, since the taper region has a varying cross-section, it must be divided into several simulation spans. The number of spans was calculated using one more convergence test, denoted in Figure 11a. Twenty five simulation spans are enough. Finally, the MMI simulation can be performed. Figure 11b represents the field distribution in the device, while Figure 11c shows the power of both output ports. Power is calculated as $|S|^2$ for the corresponding S-parameter. This power will be recalculated into an imbalance and excess loss using the formulas:

$$Imbalance = 10 \lg(P_{out1}/P_{out2})$$

$$Losses = 10 \lg(P_{out1} + P_{out2})$$

The results are presented in Figure 20.

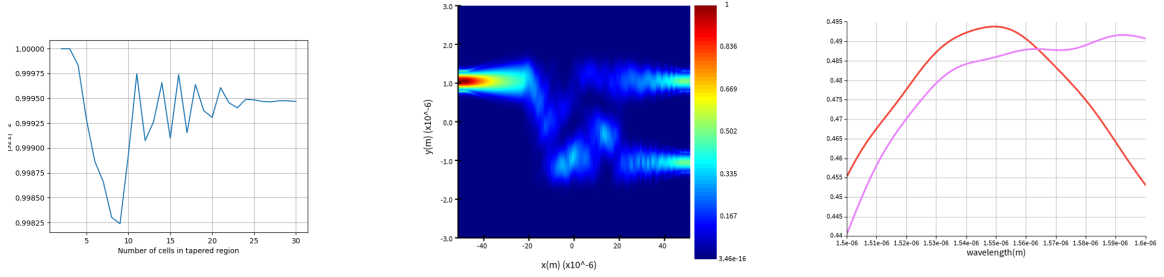


Fig. 11. (a) Number of spans convergence test for taper region, (b) MMI field profile, (c) Output powers of the MMI coupler

4. Layout for fabrication

The device layout was created in KLayout [9]. The final layout is presented in Figure 12 and contains 10 devices: 5 MZIs with different imbalances; 3 MMI couplers 2x2 (with optimal design, with wider and longer MMI part); loopback circuit for calibration; tapers with multimode short waveguide. The details of the prepared devices are listed in Table II.

Several layout constraints were incorporated to enable future automated testing. The primary considerations were as follows: first, all TE grating couplers were oriented toward the right side to streamline measurements and eliminate unnecessary chip rotations; second, the input and output grating couplers were assigned a fixed 127 μm pitch to match the spacing of the fiber array used in automated characterization. Additionally, to reduce bend losses, all curved sections of TE devices were designed with at least of 5 μm radius. For fair MMI imbalance analysis, the arms after the MMI were designed to be of an equal length.

TABLE II
List of fabricated devices and their geometries

Device	Geometry
MZI 1	$\Delta L = 47.51\mu\text{m}$
MZI 2	$\Delta L = 63.35\mu\text{m}$
MZI 3	$\Delta L = 95.02\mu\text{m}$
MZI 4	$\Delta L = 158.37\mu\text{m}$
MZI 5	$\Delta L = 190.04\mu\text{m}$
MMI 1	$L_{\text{MMI}} = 46.7\mu\text{m}, W_{\text{MMI}} = 3.6\mu\text{m}$ (optimal)
MMI 2	$L_{\text{MMI}} = 46.9\mu\text{m}, W_{\text{MMI}} = 3.6\mu\text{m}$
MMI 3	$L_{\text{MMI}} = 46.7\mu\text{m}, W_{\text{MMI}} = 3.65\mu\text{m}$
Loopback	$L = 135\mu\text{m}$
Tapers with multimode waveguide	$L_t = 25\mu\text{m}, L_{\text{mm}} = 5\mu\text{m}, W = 1.5\mu\text{m}$

5. Fabrication and measurement

5.1. Fabrication

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once

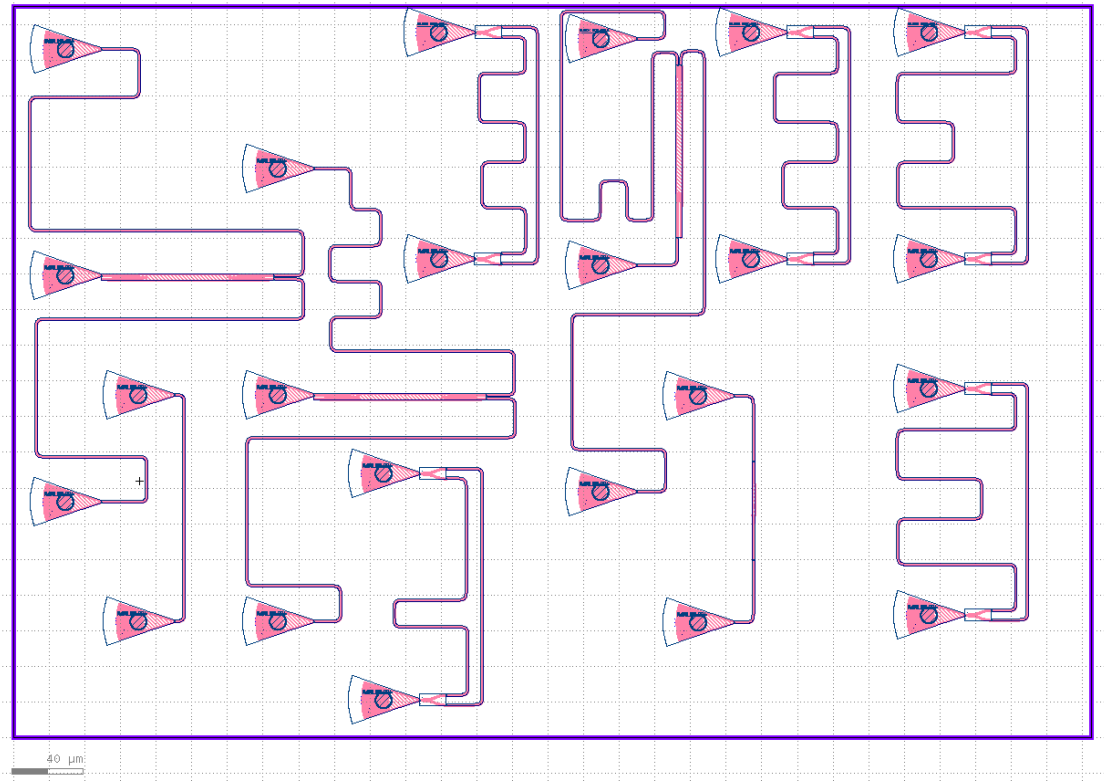


Fig. 12. Layout for fabrication

fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A $2.2\ \mu\text{m}$ oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C . Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

5.2. Measurement

To characterize the devices, a custom-built automated test setup [10] with automated control software written in Python was used [11]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [12]. A 90° rotation was used to inject light into the TM grating couplers [12]. A polarization maintaining fibre array was used to couple light in/out of the chip [13].

6. Tolerance Analysis

In this section, we focus on taking into account possible fabrication variations.

6.1. Corner Analysis

First of all, the influence of the variations in thickness and width on the effective and group indices is analyzed. We assume that the wafer thickness varies between 215.3 and 223.1 nm, and the waveguide width varies between 470 and 510 nm. For this purpose, the cross-section modeling is repeated for all corner values. Figure 13 demonstrates all possible wavelength dependencies of the effective and group indices. The Figure shows that the effective index varies from 2.37 to 2.47, while the group index – from 4.16 to 4.24.

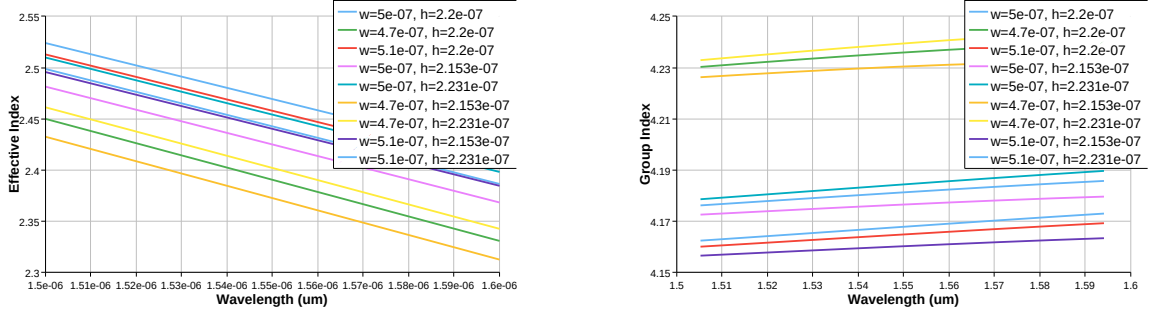


Fig. 13. Corner analysis for (a) effective index, (b) group index

7. Experimental results

7.1. MZI

This section presents experimental results and data analysis. Five MZIs have been fabricated and measured. Figure 14 demonstrates the measured raw data of the MZIs in comparison with their modeling using PDK. As could be seen, real devices have higher losses; the resonant wavelength is quite different – the longer wavelength part is weaker and noisier than expected. This could be caused, for instance, by imperfect fabrication or limited bandwidth of the source. Nevertheless, FSR seems to be close to expectations. Then the data will be processed.

To save space, all steps of the data processing will be presented using only the first MZI. The first step includes baseline correction, which allows to consider the grating coupler transfer function. Figure 15 demonstrates the polynomial fit of the grating coupler transfer function and the corrected MZI spectrum.

However, a baseline correction performs approximate results. For a more accurate analysis that takes into account the grating couplers' limited bandwidth and losses, as well as waveguide and bend losses, the data are calibrated using a loopback structure. Note, that in our case, the loopback structure has a different path length, and does not contain Y-branch splitters. Nevertheless, the result is accurate enough for further analysis. Calibration with the loopback structure is shown in Figure 16.

Then, the waveguide parameters were extracted. The calibrated with the loopback circuit MZI spectrum was used for calculating the FSR by finding peaks, see Figure 17. Thus, the group index could be found, even in the particular spectrum there were only five peaks. However, it is better to use the found group index value as an initial parameter for the MZI transmission spectrum curve fit and to adjust the effective index to match the model with the experimental data. The resulting fit was obtained and is shown in Figure 18a. Then, all waveguide parameters could be calculated. The group index obtained from the curve fit model illustrated in Figure 18b.

The measurement data analysis described above was performed for all five MZI spectra. The waveguide parameters for wavelength of 1.55 μm are provided in the table III. Figure 19 demon-

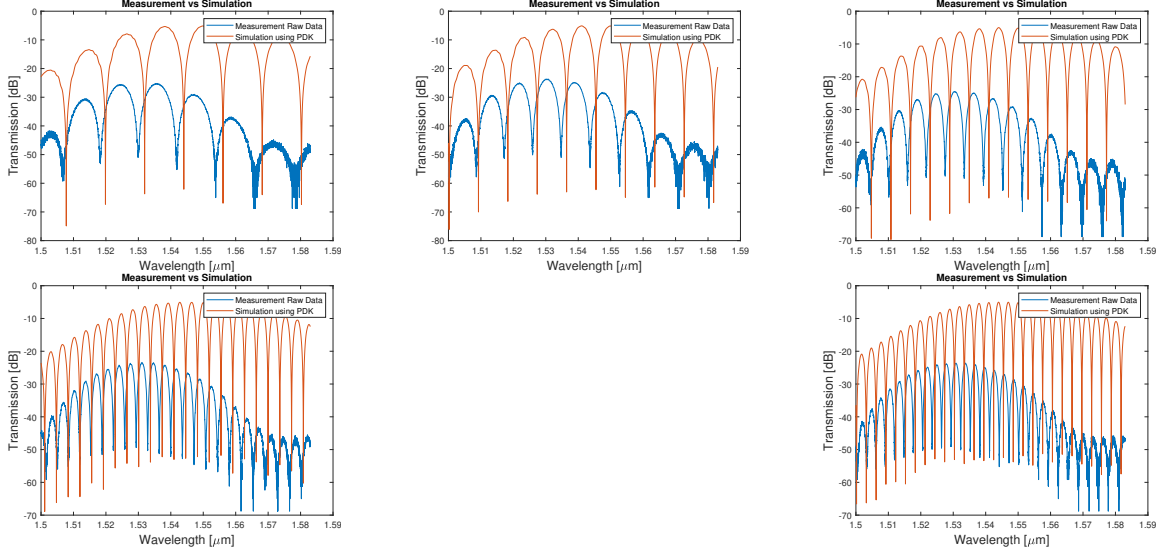


Fig. 14. The comparison of the MZIs' raw measured data with the modeling using PDK

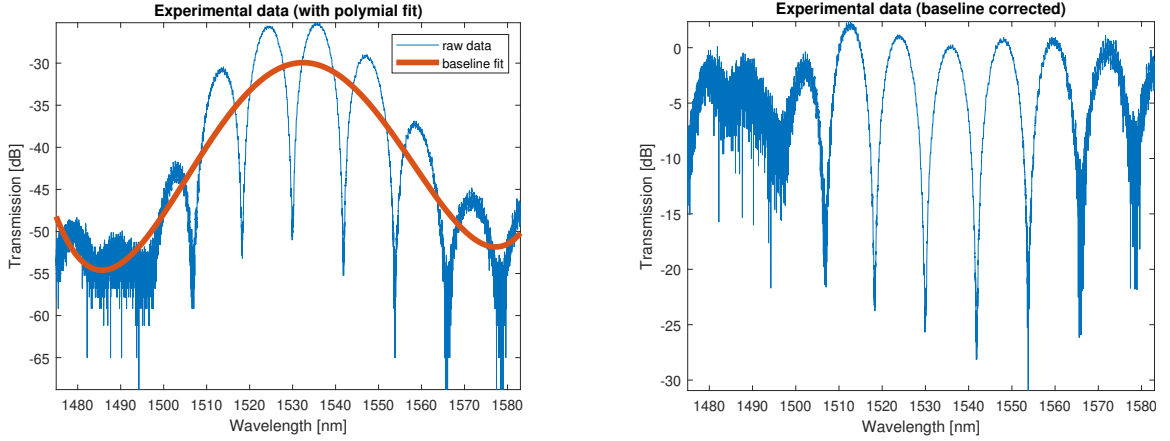


Fig. 15. Raw data and baseline correction for the first MZI at room temperature

strates comparison of the calibrated and curve fitted measurement data with circuit simulation for all five MZIs.

TABLE III
Waveguide parameters for wavelength of $1.55 \mu\text{m}$ for simulated nominal design, minimal and maximal group index corner values, and 5 measured devices

Data source	n_{eff}	$\text{textbf{bn}}_g$	D [ps/nm/km]
Waveguide simulation	2.4424	4.1811	362.15
Corner analysis with min n_g	2.4399	4.1601	259.28
Corner analysis with max n_g	2.4018	4.2391	433.73
Measured MZI 1	2.3757	4.1944	559.03
Measured MZI 2	2.3682	4.1950	380.42
Measured MZI 3	2.3770	4.1953	-17.55
Measured MZI 4	2.3755	4.1910	502.77
Measured MZI 5	2.3781	4.1910	551.31

As mentioned above, the extracted group index is close to simulated one leading to almost the same FSR. The effective index of the fabricated devices is lower than that in the simulation; thus,

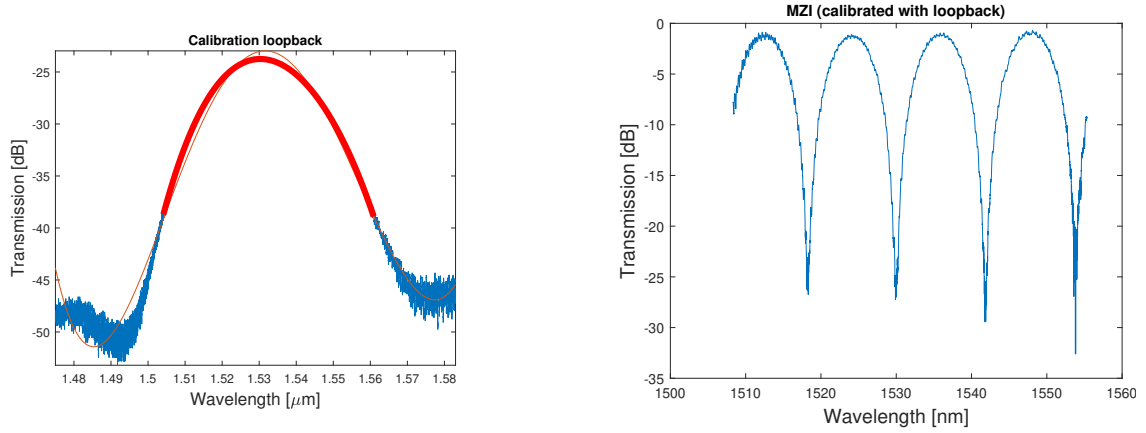


Fig. 16. (a) Loopback transmission spectrum and its polynomial fit, (b) MZI spectrum calibrated using the loopback circuit

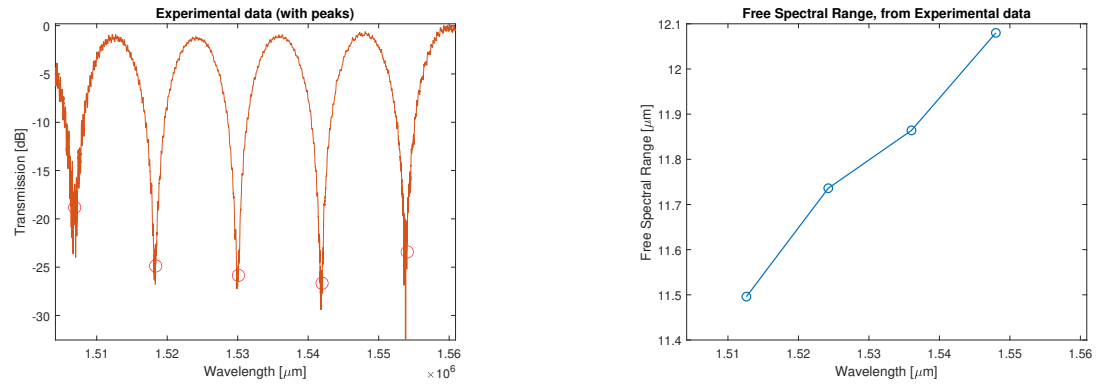


Fig. 17. (a) Peaks found in the loopback calibrated MZI spectrum (b) FSR calculated by found peaks

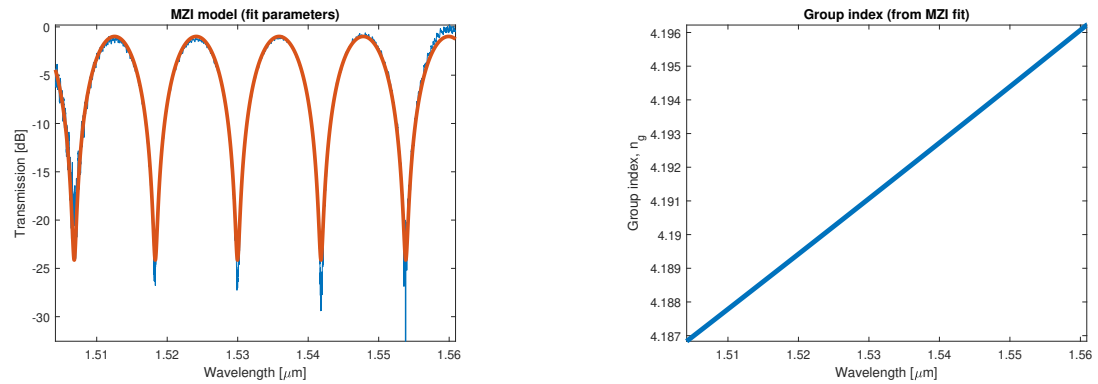


Fig. 18. (a) Curve fit of MZI spectrum (b) group index wavelength dependence from the curve fit

the resonance shift is observed. It seems that the reason for the lower effective index could be smaller waveguide sizes. The measured waveguide group index comparison with simulated corner analysis is presented in Figure 20. The measured group indices for different MZIs are close to each other and fit perfectly between two corner values. Hence, we can conclude that the corner values were chosen and simulated correctly.

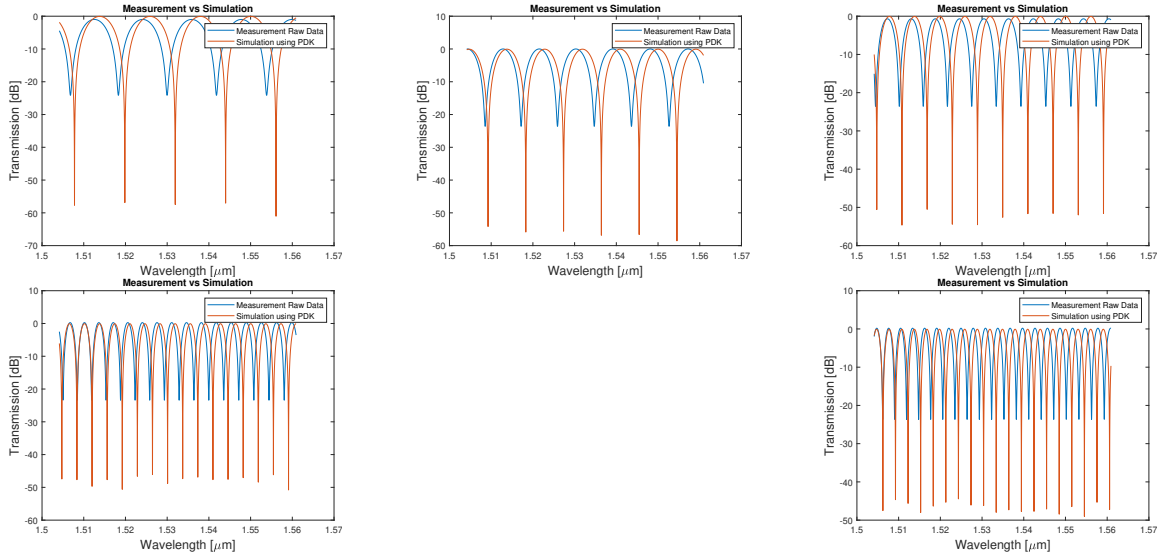


Fig. 19. The comparison of the MZIs' curve fit measured data with the modeling using standard models

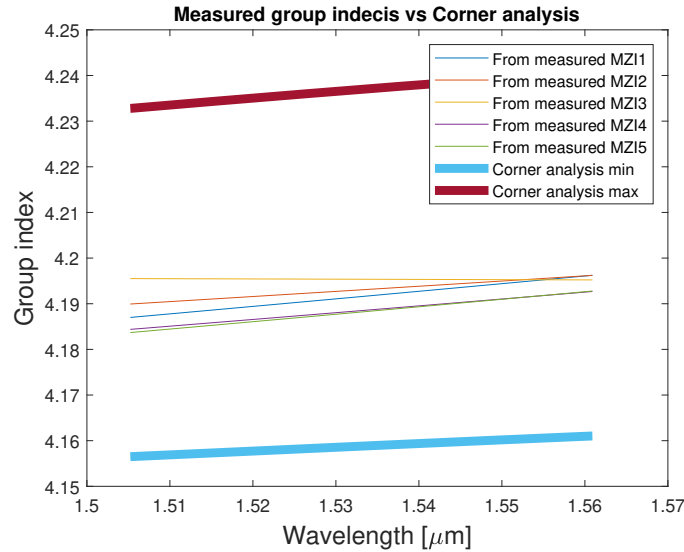


Fig. 20. Comparison of the measured group indices with the calculated minimal and maximal corner group index values

7.2. MMI

For processing MMI measurement data, the same algorithm was performed: firstly, the loopback data was fitted and used for spectral calibration. Then, the calibrated spectra of the two MMI outputs were smoothed to remove noise. After, the imbalance and excess losses were calculated. The results in comparison with the simulated data are denoted in Figure 21. The imbalance is higher for the measured device than expected; however, it is still lower than 0.7 dB. The excess loss results seem to be inaccurate since some of the spectrum has negative losses, which is not physical result. This might happen due to several reasons: quality of the Bragg gratings for the loopback and the MMI circuits could be different; the measurement of the circuits could be done with different precision. These reasons could affect since the expected losses are less than 0.5 dB and some fabrication or measurement variations could cause a difference of more than 0.5 dB.

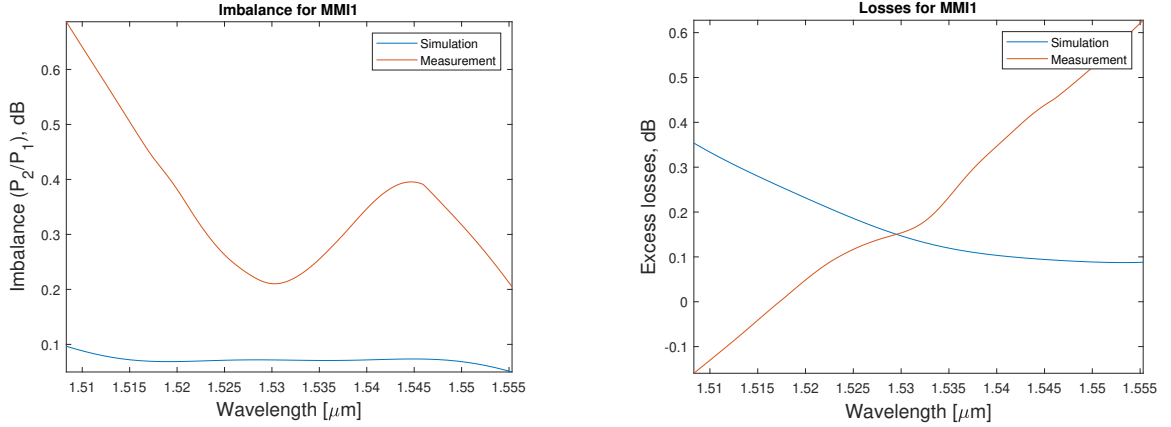


Fig. 21. (a) Imbalance of the MMI (b) Excess losses of the MMI

8. Conclusion

In this work, the full PIC design workflow was explored, beginning with element-level simulations, followed by circuit-level integration, and ultimately the layout generation required for fabrication. Within this framework, several MZIs were designed, fabricated, and experimentally characterized. A corner analysis was performed to assess the robustness of the devices to fabrication variations, and corresponding measurements from the manufactured chip were collected and evaluated.

From the experimental data, the wavelength dependence of the group refractive index was successfully extracted, and effective refractive indices were estimated. The measured effective indices were slightly lower than expected, while the group index showed good agreement with the predicted corner-based variability range.

In addition to MZI structures, three MMI couplers were designed and fabricated, one of which underwent detailed evaluation. The measured imbalance was marginally higher than the predicted value, and loss estimation proved challenging due to limitations associated with the loopback structure. Consequently, it is difficult to draw definitive conclusions regarding measurement-to-simulation agreement. However, the results indicate that the splitter functions as intended, and further corner analysis should be conducted to quantify the spread and tolerances of its performance metrics. This work may serve as the basis for future refinement and verification.

Overall, the course provided experience across all stages of PIC development – from theoretical background and component-level modeling through circuit design, layout preparation, fabrication considerations, and final measurement. The acquired results establish a solid foundation for deeper performance analysis and process optimization in subsequent work. The acquired knowledge, skills, and preliminary results are expected to form a strong foundation for more advanced scientific studies, refined PIC designs, and further development of integrated photonic technologies.

9. Acknowledgments

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Omid Esmaeeli performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Python, and KLayout for the design software.

10. References

1. Shekhar S, Bogaerts W, Chrostowski L, Bowers JE, Hochberg M, Soref R, et al. Roadmapping the next generation of silicon photonics. *Nat Commun.* 2024;15(1):751
2. Fu T, Zhang J, Sun R, Huang Y, Xu W, Yang S, et al. Optical neural networks: progress and challenges. *Light Sci Appl.* 2024.
3. Chrostowski L, Hochberg ME. *Silicon photonics design*. Cambridge: Cambridge university press; 2015.
4. Soldano LB, Pennings ECM. Optical multi-mode interference devices based on self-imaging: principles and applications. *J Lightwave Technol.* 1995 Apr;13(4):615–27.
5. Yang T, Zhang E, Zhang S, Yang S, Yang X, Zhang L. Low cross talk, low P silicon optical switch based on highly balanced couplers and folded phase shifters. *Opt Lett.* 2025 Jan 1;50(1):101.
6. Lumerical MODE software. <https://www.lumerical.com/tcad-products/mode/>
7. Components library, SiEPIC, University of Washington. <http://goo.gl/B9IRn8>
8. Lumerical INTERCONNECT software. <https://www.lumerical.com/tcad-products/interconnect/>
9. KLayout software. <http://www.klayout.de/>.
10. <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.
11. <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.
12. Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", *Optics Express* Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652
13. www.plcconnections.com, PLC Connections, Columbus OH, USA.